6. Pengolahan Data

1. Data Tunggal

2. Sumber Tegangan

1). $V = 3V$ (DC)

$\Delta V = \frac{1}{2} NST = \frac{1}{2} 0{,}001$

$= 0{,}0005\,V$

$KSR = \frac{\Delta V}{V} \times 100\% = \frac{0{,}0005}{3} \times 100\%$

$= 0{,}016\%$ (4AP)

$(V \pm \Delta V) = (3{,}000 \pm 0{,}0005000)V$

2) $V = 6V$ (DC)

$\Delta V = \frac{1}{2} NST = \frac{1}{2} \cdot 0{,}001$

$= 0{,}0005V$

$KSR = \frac{\Delta V}{V} \times 100\% = \frac{0{,}0005}{6} \times 100\%$

$= 0{,}008\overline{3}\%$ (4AP)

$(V \pm \Delta V) = (6{,}000 \pm 0{,}0005000)V$

3). $V = 9V$

$\Delta V = \frac{1}{2} NST = \frac{1}{2} 0{,}001$

$= 0{,}0005V$

$KSR = \frac{\Delta V}{V} \times 100\% = \frac{0{,}0005}{9} \times 100\%$

$= 0{,}005\%$ (4AP)

$(V \pm \Delta V) = (9{,}000 \pm 0{,}0005000)V$

b. Arus

1). Arus 3V (DC)

$I = 0{,}0049\,A$

$\Delta I = \frac{1}{2} NST = \frac{1}{2} \times 0{,}001 = 0{,}0005\,A$

$KSR = \frac{\Delta I}{I} \times 100\% = \frac{0{,}0005}{0{,}0049} \times 100\%$

$= 10{,}20\%$ (Error)

$(I \pm \Delta I) = (0{,}0049 \pm 0{,}0005)\,A$

2). Arus 6V (DC)

$I = 0{,}00084\,A$

$\Delta I = 0{,}0005\,A$

$KSR = \frac{\Delta I}{I} \times 100\% = 4{,}61\%$ (2AP

$(I \pm \Delta I) =$ ~~to~~ $(0{,}011 \pm 0{,}00050)A$

3. Arus 9V (DC)

$I = 0{,}0459\,A$

$\Delta I = 0{,}0005A$

$KSR = \frac{\Delta I}{I} \times 100\% = \frac{0{,}0005}{0{,}0459} \times 100\%$

$= 1{,}1\%$ (2AP)

$(I \pm \Delta I) = (0{,}046 \pm 0{,}00050)A$

c. Menghitung Tegangan DC 3V

1). $V_{ab} = 3{,}259V$, $\Delta V = 0{,}0005V$

$KSR = \frac{\Delta V}{V_{ab}} \times 100\% = 0{,}015\%$ (4AP)

$(V_{ab} \pm \Delta V) = (3{,}259 \pm 0{,}0005000)V$

2). $V_{ac} = 0{,}0321\,V$, $\Delta V = 0{,}0005V$

$KSR = \frac{\Delta V}{V_{ab}} \times 100\% = 1{,}6\%$ (2AP)

$(V_{ac} \pm \Delta V) = (0{,}032 \pm 0{,}00050)\,V$

3). $V_{ad} = 0{,}0256V$, $\Delta V = 0{,}0005V$

$KSR = \frac{\Delta V}{V_{ad}} \times 100\% = 1{,}9\%$ (2AP)

$(V_{ad} \pm \Delta V) = (0{,}026 \pm 0{,}00050)V$

4). $V_{ae} = 3{,}113V$, $\Delta V = 0{,}0005V$

$KSR = \frac{\Delta V}{V_{ae}} \times 100\% = 0{,}01\%$ (4AP)

$(V_{ae} \pm \Delta V) = (3{,}113 \pm 0{,}0005000)$ V

5). $V_{bc} = 1{,}371V$, $\Delta V = 0{,}0005V$

$KSR = \frac{\Delta V}{V_{bc}} \times 100\% = 0{,}04\%$ (4AP)

$(V_{bc} \pm \Delta V) = (1{,}371 \pm 0{,}0005000)V$

6). $V_{bd} = 6{,}11V$, $\Delta V = 0{,}0005V$

$KSR = \frac{\Delta V}{V_{bd}} \times 100\% = 0{,}008\%$ (4AP)

$(V_{bd} \pm \Delta V) = (6{,}110 \pm 0{,}0005000)V$

7). $V_{be} = 6{,}27V$, $\Delta V = 0{,}0005V$

$KSR = \frac{\Delta V}{V_{be}} \times 100\% = 0{,}008\%$ (4AP)

$(V_{be} \pm \Delta V) = (6{,}270 \pm 0{,}0005000)V$

8). $V_{cd} = 0{,}3705V$, $\Delta V = 0{,}0005V$

$KSR = \frac{\Delta V}{V_{cd}} \times 100\% = 0{,}13\%$ (3AP)

$(V_{cd} \pm \Delta V) = (0{,}370 \pm 0{,}000500)V$

9). $V_{ce} = 0{,}3945V$, $\Delta V = 0{,}0005V$

$KSR = \frac{\Delta V}{V_{ce}} \times 100\% = 0{,}13\%$ (3AP)

$(V_{ce} \pm \Delta V) = (0{,}394 \pm 0{,}000500)V$

10). $V_{de} = 0{,}8V$, $\Delta V = 0{,}0005V$

$KSR = \frac{\Delta V}{V_{de}} \times 100\% = 0{,}06\%$ (4AP)

$(V_{de} \pm \Delta V) = (0{,}800 \pm 0{,}0005000)V$

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d. Menghitung Tegangan DC 6V

1). $V_{ab} = 0,61V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{ab}} \times 100\% = 0,082\%$ (4AP)

$(V_{ab} \pm \Delta V) = (0,6100 \pm 0,00050000)V$

2). $V_{ac} = 1,77V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{ac}} \times 100\% = 0,028\%$ (4AP)

$(V_{ac} \pm \Delta V) = (1,770 \pm 0,00050000)V$

3). $V_{ad} = 4,18V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{ad}} \times 100\% = \frac{0,0005}{4,18} = 0,012\%$ (4AP)

$(V_{ad} \pm \Delta V) = (4,180 \pm 0,00050000)V$

4). $V_{ae} = 6,15V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{ae}} \times 100\% = 0,008\%$ (4AP)

$(V_{ae} \pm \Delta V) = (6,150 \pm 0,00050000)V$

5). $V_{bc} = 1,088V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{bc}} \times 100\% = 0,046\%$ (4AP)

$(V_{bc} \pm \Delta V) = (1,088 \pm 0,00050000)V$

6) $V_{bd} = 3,484V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{bd}} \times 100\% = 0,014\%$ (4AP)

$(V_{bd} \pm \Delta V) = (3,484 \pm 0,00050000)V$

7). $V_{be} = 5,46V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{be}} \times 100\% = 0,009\%$ (4AP)

$(V_{be} \pm \Delta V) = (5,460 \pm 0,00050000)V$

8). $V_{cd} = 2,396V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{cd}} \times 100\% = 0,021\%$ (4AP)

$(V_{cd} \pm \Delta V) = (2,396 \pm 0,00050000)V$

9). $V_{ce} = 4,37V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{ce}} \times 100\% = 0,011\%$ (4AP)

$(V_{ce} \pm \Delta V) = (4,370 \pm 0,00050000)V$

10). $V_{de} = 1,968V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{de}} \times 100\% = 0,025\%$ (4AP)

$(V_{de} \pm \Delta V) = (1,968 \pm 0,00050000)V$

e. Menghitung Tegangan DC 9V

1). $V_{ab} = 0,71V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{ab}} \times 100\% = 0,07\%$ (4AP)

$(V_{ab} \pm \Delta V) = (0,7100 \pm 0,00050000)V$

2). $V_{ac} = 2,405V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{ac}} \times 100\% = 0,02\%$ (4AP)

$(V_{ac} \pm \Delta V) = (2,405 \pm 0,00050000)V$

3). $V_{ad} = 6,15V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{ad}} \times 100\% = 0,008\%$ (4AP)

$(V_{ad} \pm \Delta V) = (6,150 \pm 0,00050000)V$

4). $V_{ae} = 9,22V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{ae}} \times 100\% = 0,005\%$ (4AP)

$(V_{ae} \pm \Delta V) = (9,220 \pm 0,00050000)V$

5) $V_{bc} = 1,695V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{bc}} \times 100\% = 0,029\%$ (4AP)

$(V_{bc} \pm \Delta V) = (1,695 \pm 0,00050000)V$

6). $V_{bd} = 5,45V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{bd}} \times 100\% = 0,009\%$ (4AP)

$(V_{bd} \pm \Delta V) = (5,450 \pm 0,00050000)V$

7). $V_{be} = 8,51V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{be}} \times 100\% = 0,006\%$ (4AP)

$(V_{be} \pm \Delta V) = (8,510 \pm 0,00050000)V$

8). $V_{cd} = 3,74V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{cd}} \times 100\% = 0,013\%$ (4AP)

$(V_{cd} \pm \Delta V) = (3,740 \pm 0,00050000)V$

9). $V_{ce} = 6,81V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{ce}} \times 100\% = 0,007\%$ (4AP)

$(V_{ce} \pm \Delta V) = (6,810 \pm 0,00050000)V$

10). $V_{de} = 3,608V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{de}} \times 100\% = 0,01\%$ (4AP)

$(V_{de} \pm \Delta V) = (3,608 \pm 0,00050000)V$

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f. Menghitung Sumber Tegangan AC 3 V_{pp}

1). $V_{ab} = 0,9440V$, $\Delta V = 0,0005V$, $KSR = \frac{V_{ab}}{\Delta V} \times 100\% = 0,53\%$ (3AP)

$(V_{ab} \pm \Delta V) = (0,944 \pm 0,00050)V$

2). $V_{ac} = 0,988V$, $\Delta V = 0,0005V$, $KSR = \frac{V_{ac}}{\Delta V} \times 100\% = 0,51\%$ (3AP)

$(V_{ac} \pm \Delta V) = (0,988 \pm 0,00050)V$

3). $V_{ad} = 1,05V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{ad}} \times 100\% = 0,48\%$ (3AP)

$(V_{ad} \pm \Delta V) = (1,05 \pm 0,00050)V$

4). $V_{ae} = 1,1V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{ae}} \times 100\% = 0,45\%$ (3AP)

$(V_{ae} \pm \Delta V) = (1,10 \pm 0,00050)V$

5). $V_{bc} = 0,066V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{bc}} \times 100\% = 7,6\%$ (2AP)

$(V_{bc} \pm \Delta V) = (0,066 \pm 0,00050)V$

6). $V_{bd} = 0,16V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{bd}} \times 100\% = 3,1\%$ (2AP)

$(V_{bd} \pm \Delta V) = (0,16 \pm 0,00050)V$

7). $V_{be} = 0,212V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{be}} \times 100\% = 2,35\%$ (2AP)

8). $V_{cd} = 0,110V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{cd}} \times 100\% = 4,5\%$ (2AP)

$(V_{cd} \pm \Delta V) = (0,11 \pm 0,00050)V$

9). $V_{ce} = 0,169V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{ce}} \times 100\% = 3\%$ (2AP)

$(V_{ce} \pm \Delta V) = (0,17 \pm 0,00050)V$

10) $V_{de} = 0,076V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{de}} \times 100\% = 6,6\%$ (2AP)

$(V_{de} \pm \Delta V) = (0,076 \pm 0,00050)V$

g. Menghitung Tegangan AC 6 V_{pp}

i). $V_{ab} = 1,75V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{ab}} \times 100\% = 0,28\%$ (3AP)

$(V_{ab} \pm \Delta V) = (1,75 \pm 0,00050)V$

2). $V_{ac} = 1,874$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{ac}} \times 100\% = 0,27\%$ (3AP)

$(V_{ac} \pm \Delta V) = (1,87 \pm 0,00050)V$

3). $V_{ad} = 2,06V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{ad}} \times 100\% = 0,24\%$ (3AP)

$(V_{ad} \pm \Delta V) = (2,06 \pm 0,00050)V$

4). $V_{ae} = 2,61V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{ae}} \times 100\% = 0,19\%$ (3AP)

$(V_{ae} \pm \Delta V) = (2,61 \pm 0,00050)V$

5). $V_{bc} = 0,174V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{bc}} \times 100\% = 2,9\%$ (2AP)

$(0,17 \pm 0,00050)V = (V_{bc} \pm \Delta V)V$

6). $V_{bd} = 0,424V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{bd}} \times 100\% = 1,2\%$ (2AP)

$(V_{bd} \pm \Delta V) = (0,42 \pm 0,0050)V$

7). $V_{be} = 0,557V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{be}} \times 100\% = 0,89\%$ (3AP)

$(V_{be} \pm \Delta V) = (0,557 \pm 0,00050)V$

8). $V_{cd} = 0,293V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{cd}} \times 100\% = 1,71\%$ (2AP)

$(V_{cd} \pm \Delta V) = (0,29 \pm 0,0050)V$

9). $V_{ce} = 0,44V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{ce}} \times 100\% = 1,13\%$ (2AP)

$(V_{ce} \pm \Delta V) = (0,44 \pm 0,00050)V$

10). $V_{de} = 0,198V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{de}} \times 100\% = 2,51\%$ (2AP)

$(V_{de} \pm \Delta V) = (0,20 \pm 0,00050)V$

h. Menghitung Tegangan AC 9 V_{pp}

1) $V_{ab} = 2,57V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{ab}} \times 100\% = 0,19\%$ (3AP)

$(V_{ab} \pm \Delta V) = (2,57 \pm 0,00050)V$

2). $V_{ac} = 2,78V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{ac}} \times 100\% = 0,18\%$ (3AP)

$(V_{ac} \pm \Delta V) = (2,78 \pm 0,00050)V$

3). $V_{ad} = 3,207V$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{ad}} \times 100\% = 0,16\%$ (3AP)

$(V_{ad} \pm \Delta V) = (3,21 \pm 0,00050)V$

4). $V_{ae} \pm 3,26$, $\Delta V = 0,0005V$, $KSR = \frac{\Delta V}{V_{ae}} \times 100\% = 0,15\%$ (3AP)

$(V_{ae} \pm \Delta V) = (3,26 \pm 0,00050)V$

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5). $V_{bc} = 0{,}278V$, $\Delta V = 0{,}005V$ $\quad KSR = \frac{\Delta V}{V_{bc}} \times 100\% = 1{,}8\%$ (2AP)

$(V_{bc} \pm \Delta V) = (0{,}28 \pm 0{,}0050)V$

6). $V_{bd} = 0{,}700V$, $\Delta V = 0{,}005V$ $\quad KSR = \frac{\Delta V}{V_{bd}} \times 100\% = 0{,}71\%$ (3AP)

$(V_{bd} \pm \Delta V) = (0{,}700 \pm 0{,}00500)V$

7). $V_{be} = 0{,}919V$, $\Delta V = 0{,}005V$ $\quad KSR = \frac{\Delta V}{V_{be}} \times 100\% = 0{,}54\%$ (3AP)

$(V_{be} \pm \Delta V) = (0{,}919 \pm 0{,}00500)V$

8). $V_{cd} = 0{,}502V$, $\Delta V = 0{,}005V$ $\quad KSR = \frac{\Delta V}{V_{cd}} \times 100\% = 0{,}99\%$ (3AP)

$(V_{cd} \pm \Delta V) = (0{,}502 \pm 0{,}00500)V$

9). $V_{ce} = 2{,}09V$, $\Delta V = 0{,}005V$ $\quad KSR = \frac{\Delta V}{V_{ce}} \times 100\% = 0{,}24\%$ (3AP)

$(V_{ce} \pm \Delta V) = (2{,}09 \pm 0{,}00500)V$

10). $V_{de} = 0{,}329V$, $\Delta V = 0{,}005V$ $\quad KSR = \frac{\Delta V}{V_{de}} \times 100\% = 1{,}52\%$ (2AP)

$(V_{de} \pm \Delta V) = (0{,}33 \pm 0{,}005)$

2. Data majemuk

↳ Tidak ada data majemuk

H. Perhitungan & Analisis

1. Perhitungan

↳ Tidak ada perhitungan

2. Analisis